

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

PHYSICS ORDINARY LEVEL

6118/2

PAPER 2

1 hour 30 minutes

Marks 80

2024

Additional Materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces provided on top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You will lose marks if you do not show your working or if you do not use appropriate units.
- Take the weight of 1 kg to be 10 N (i.e acceleration of free fall $g = 10 \text{ m/s}^2$).
- Periodic Table is printed on page 14.

For Examiner's Use

1	
2	
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Total	

Marker	
Checker	

This document consists of **14** printed pages and **2** blank pages.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 (a) Complete Table 1.1 to state whether the physical quantities are scalars or vectors.

Table 1.1

physical quantity	scalar or vector
speed	
velocity	
distance	
force	
kinetic energy	

[3]

- (b) Fig. 1.1 shows how the speed of a car varies over a short time. Different parts of the journey are labelled A, B, C, D and E.

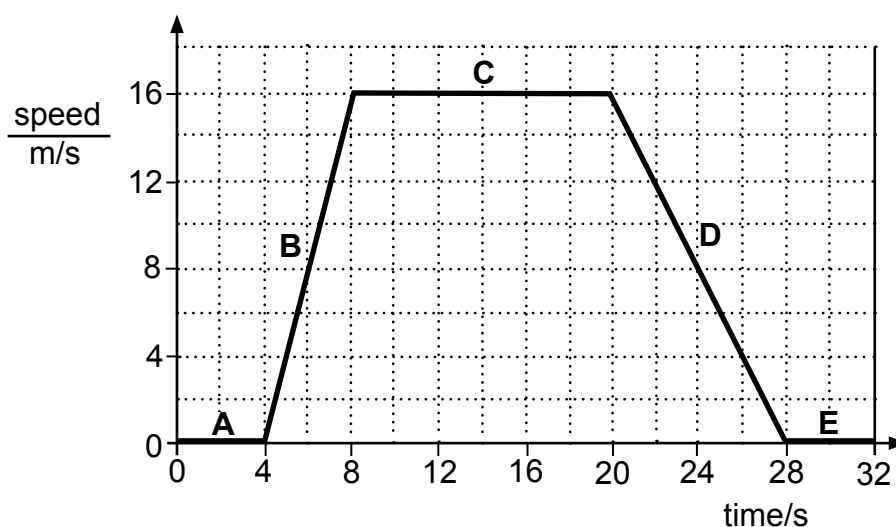


Fig. 1.1

- (i) State a part of the graph that shows the car at rest.

..... [1]

- (ii) State a part of the graph that shows the car moving with constant speed.

..... [1]

- (iii) During part of the journey the speed of the car increases.

Calculate the acceleration of the car and state the unit.

acceleration = unit = [3]

- (c) Another car accelerates, from rest, starting at time 0 s. This car has a constant acceleration. Its speed at 20 s is 10 m/s.

On Fig. 1.1, draw a line to show this motion.

[2]

- (d) Describe, using Fig. 1.1, how you can determine which car has the greater acceleration.

.....

.....

[1]

- (e) A particle moves with a speed of 4 m/s for 20 s. It again moves with a speed of 6 m/s for another 20 s and finally moves with a speed of 8 m/s for the next 10 s.

Calculate the average speed of the particle.

average speed =m/s [3]

[14]

- 2 A man uses a metal bar to remove an iron nail from a piece of wood, as shown in Fig. 2.1.

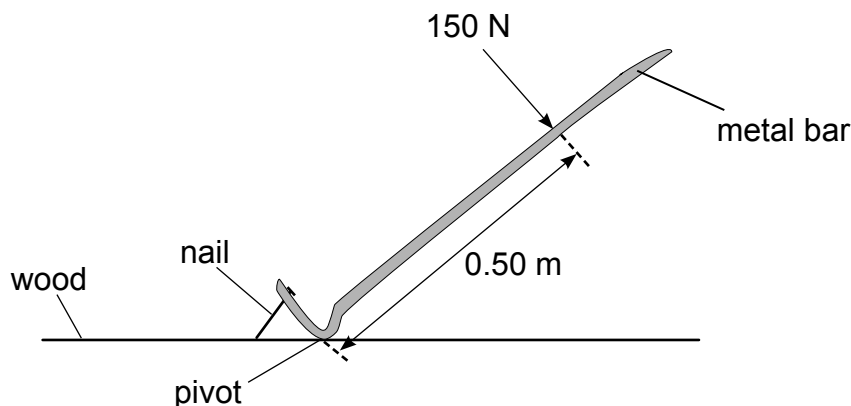


Fig. 2.1

- (a) (i) The man applies a force of 150 N perpendicular to the bar and at a distance of 0.50 m from the pivot.

Calculate the moment of this force about the pivot. State the unit.

moment = unit = [3]

- (ii) The force applied by the man produces a turning effect (moment) about the pivot.

State **two** other examples of using the turning effect of a force.

..... [2]

- (b) The man tries to use the metal bar to remove another nail from the piece of wood. The turning effect produced is not enough to remove this nail from the piece of wood.

Describe how the man can increase the turning effect without increasing the force.

..... [1]

- (c) State **one** condition for a system to be in equilibrium.

..... [1]

[7]

- 3 A certain substance is in the solid state at a temperature of -36°C . It is heated at a constant rate for 28 minutes. The record of its temperature is given in Table 3.1.

Table 3.1

time/min	0	1	2	6	10	14	18	22	24	26	28
temperature/ $^{\circ}\text{C}$	-36	-16	-9	-9	-9	-9	32	75	101	121	121

- (a) Define *latent heat*.

.....
 [2]

- (b) State a time in Table 3.1 at which the energy is being supplied as latent heat of fusion.

..... [1]

- (c) Explain the energy changes undergone by the molecules of a substance during the period when latent heat of vaporisation is being supplied.

.....

 [2]

- (d) (i) The rate of heating is 2.0 kW.

Calculate how much energy is supplied to the substance during the period 18 to 22 minutes.

energy supplied =J [2]

- (ii) The specific heat capacity of the substance is $1760 \text{ J / (kg } ^{\circ}\text{C)}$.

Use the information in Table 3.1 for the period 18 to 22 minutes to calculate the mass of the substance being heated.

mass =kg [3]

[10]

- 4 Fig. 4.1 shows circular wavefronts produced at the centre of a circular ripple tank.

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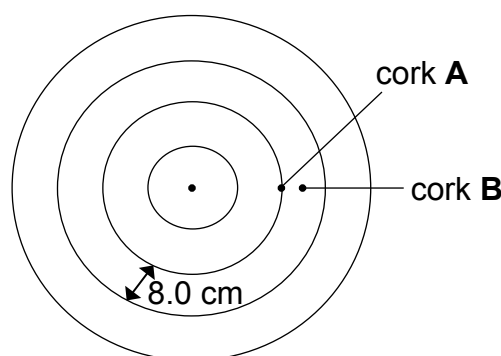


Fig. 4.1

Two corks, **A** and **B**, float on the water in the ripple tank. They move up and down on the surface of the water as the wave passes. The wavelength of the wave is 8.0 cm.

Fig. 4.2 shows how the displacement of **A** varies with time.

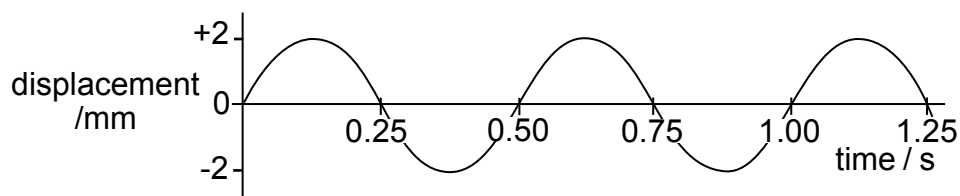


Fig. 4.2

- (a) State the amplitude of the vibrations of **A** as the wave passes.

amplitude = mm [1]

- (b) The horizontal distance between **A** and **B** is half the wavelength of the wave. On Fig. 4.2, sketch a graph to show how the displacement of **B** varies with time. [2]

- (c) (i) Use Fig. 4.2 to determine the frequency of the wave.

frequency = Hz [2]

- (ii) The distance between the centre of the ripple tank and its edge is 40 cm. Determine the time taken by a wave front to travel from the centre of the tank to the edge.

time = s [2]

- (d) A ray of light refracts as it travels from air into glass, as shown in Fig.4.3.

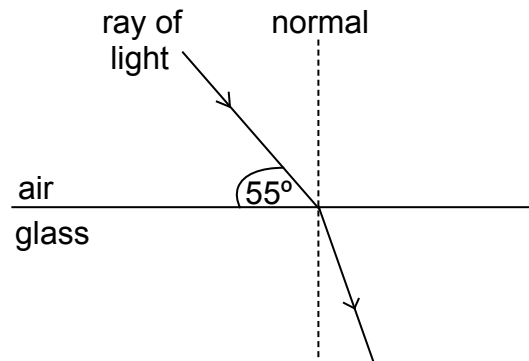


Fig. 4.3

- (i) Determine the angle of incidence of the ray of light on the glass.

angle of incidence =° [1]

- (ii) Glass has a refractive index of 1.5.

Calculate the angle of refraction of the light in the glass.

angle of refraction =° [2]

- (e) Fig. 4.4 represents some wave fronts as they pass from deep water to shallow water.

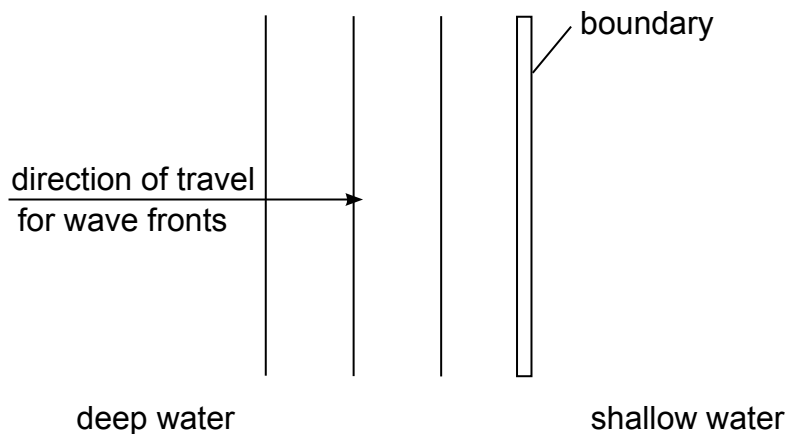


Fig. 4.4

- (i) Define the term *wave front*.

.....
.....

[1]

- (ii) On Fig.4.4, draw **three** wave fronts to show what happens to the wave fronts in the shallow water.

[3]

[14]

- 5 Fig. 5.1 shows some parts of the electromagnetic spectrum.

radio waves	Q	infrared waves	visible light	ultraviolet waves	X-rays	γ -rays
-------------	----------	----------------	---------------	-------------------	--------	----------------

Fig. 5.1

- (a) State the name of part **Q**.

..... [1]

- (b) The speed of visible light waves in a vacuum is 3.0×10^8 m/s.

State the value of the speed of infrared waves in a vacuum.

speed =m/s [1]

- (c) Give an example of the part of the electromagnetic spectrum which has a shorter wavelength than that of visible light.

..... [1]

- (d) X-rays and γ -rays are used in hospitals.

- (i) Describe **one** medical use for X-rays and **one** medical use for γ -rays.

X-rays.....

.....

γ -rays

..... [2]

- (ii) State **two** reasons why γ -rays are dangerous to living organisms.

1

.....

2

..... [2]

[7]

- 6 Fig. 6.1 shows a circuit diagram.

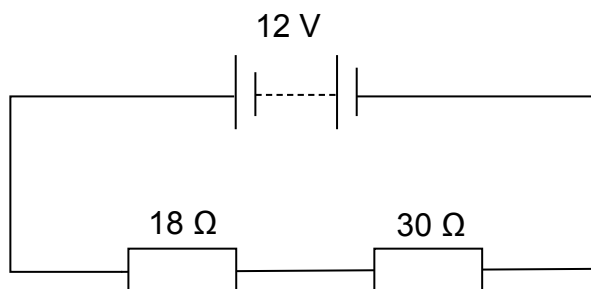


Fig. 6.1

The electromotive force (e.m.f.) of the battery is 12 V.

- (a) State what is meant by *electromotive force*.

.....

[2]

- (b) Calculate the current in the battery.

current = A

[2]

- (c) Calculate the potential difference (p.d.) across the 18 Ω resistor.

p.d. = V

[1]

- (d) Calculate the power produced in the 18 Ω resistor.

power = W

[2]

- (e) The resistors obey Ohm's law.

State Ohm's law and discuss the temperature limitation on this law.

.....

.....

.....

[2]

- (f) Describe how the resistance of a wire would change, if it replaced by

(i) a wire of the same material and length but double the diameter,

.....

[1]

(ii) a wire of the same length and diameter but hotter.

.....

[1]

[11]

- 7 Fig. 7.1. shows a simplified diagram of an Alternating current (a.c) generator connected to a $25\ \Omega$ resistor. The coil rotates anticlockwise.

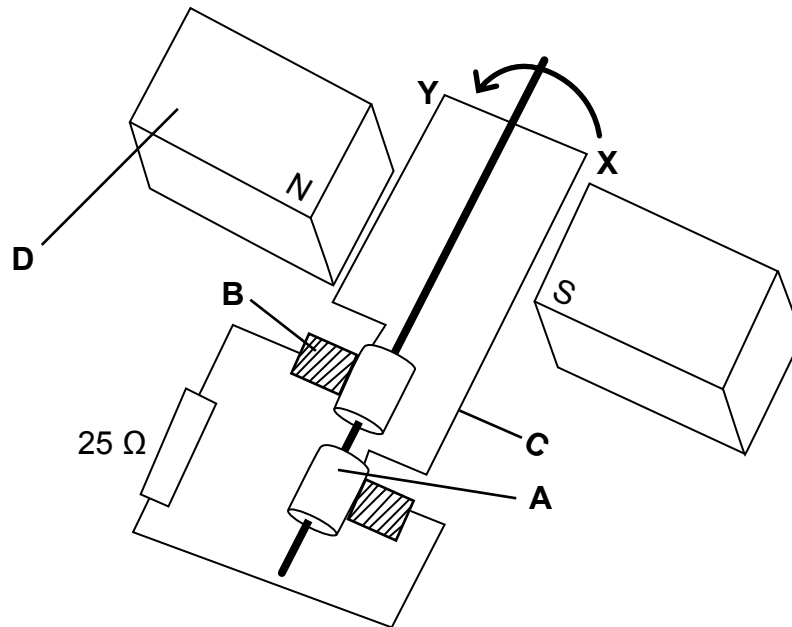


Fig. 7.1

- (a) State which component **A**, **B**, **C** or **D** distinguishes this generator from a direct current (d.c.) generator.

answer =

[1]

- (b) Name component **A** on Fig. 7.1.

.....

[1]

- (c) In which direction will the induced current flow in section **XY** of the coil?

.....

[1]

- (d) Fig. 7.2 shows the output voltage of the generator for one cycle of rotation of the coil.

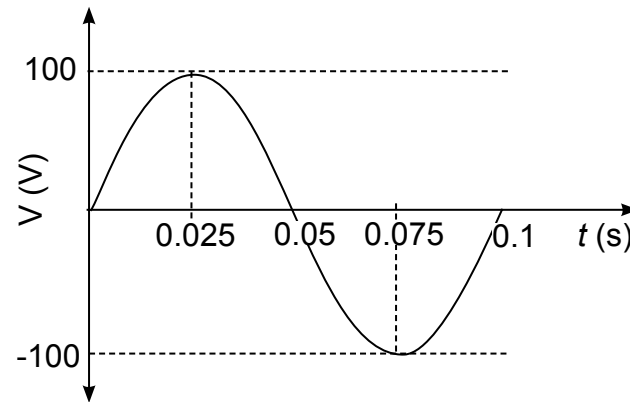


Fig. 7.2

The speed of rotation of the coil in the generator is now doubled.
Sketch the graph of output voltage versus time for 0.1 s on Fig. 7.3.

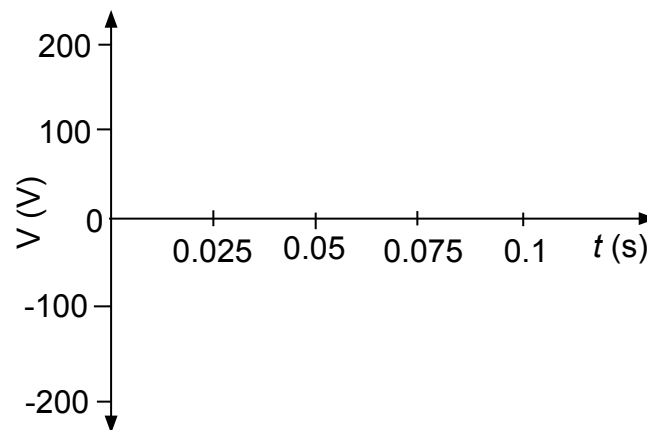


Fig. 7.3

[3]

[6]

8 Plutonium-238 is a radioactive isotope that decays by emitting alpha particles.

(a) What is meant by *radioactive decay*?

.....
..... [2]

(b) The nuclide notation for plutonium-238 is $^{238}_{94}\text{Pu}$.

In the space below, write the equations to show the decay of plutonium-238.

.....
..... [2]

(c) Alpha-particles, beta-particles and gamma-rays all cause ionisation.

The ionisation of substances in the human body can be hazardous.

(i) State how the ionising effect of beta-particles compares with the ionising effect of alpha-particles,

..... [1]

the ionising effect of gamma-rays.

..... [1]

(ii) State **two** safety precautions that are taken when handling radioactive materials.

.....
..... [2]

(d) Another radioactive isotope has a half-life of 6.0 years. A sample of this isotope has a mass of 12 mg.

Calculate the mass of this isotope that remains in the sample after 18 years.

mass remaining = mg [3]

[11]

DATA SHEET																				
The Periodic Table of the Elements																				
Group																				
I	II											III	IV	V	VI	VII	0			
		1 H Hydrogen 1																4 He Helium 2		
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10			
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18			
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36			
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	101 Ru Ruthenium 44	106 Pd Palladium 46	103 Rh Rhodium 45	106 Ag Silver 47	108 Cd Cadmium 48	112 In Indium 49	115 Sn Tin 50	119 Sb Antimony 51	122 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54				
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83						
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																		
*58 - 71 Lanthanoid series †90 - 103 Actinoid series																				
		140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71						
		232 Th Thorium 90	238 Pa Protactinium 91	238 U Uranium 92	238 Np Neptunium 93	238 Pu Plutonium 94	238 Am Americium 95	238 Cm Curium 96	238 Bk Berkelium 97	238 Cf Californium 98	238 Es Einsteinium 99	238 Fm Fermium 100	238 Md Mendelevium 101	238 No Nobelium 102	238 Lr Lawrencium 103					
Key		a	X	a = relative atomic mass X = atomic symbol b = proton (atomic) number																
		b																		

Key

a

X

b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

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